/*THIS FILE CONTAINS MY brainstorm for the xFactor calculator.
AUTHOR: A. G. Reynolds
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Methods used to factor:

GCF
difference of squares = a^2-b^2 = (a-b)(a+b)
sum/difference of cubes
quadratic = what adds to the middle term and multiplies to the third term

these all depend on the number of terms in the equation, which I have not identified… + what of prime or irreducible polynomials? find a way to identify those... root theorem?

Root & factor theorem:

if (r) is a root of a polynomial (f(x)), then (f(r)=0).
(x-r) is a factor of (f(x)) if (f(r)=0).

So, to factor a polynomial equation, the equation is set equal to zero and find the value of x, or the root. then, this root value is used to construct the factor.

assume all input equations are already equal to zero and use xsolve to get the root.
xsolve must expand to handle higher degrees. Since the exponent is what determines degree, the operator function can be updated such that when the ^ char is found directly after a variable, the coefficient is extracted.

must prevent broken program for plain digit squared. and squared expressions. $(x+2)^2$ is not a coefficient, nor 2^2. However $(x+2)^2$ is a binomial and should return (x+2)(x+2).

This is WAY more complex than I want it to be, honestly,
but I have gotten this far, so backing out now would not be very impressive…

If I can finish the logic, I can write the code! Never surrender to Simplicity!

anyways…so the rule needs to be:

if the current char = ^

if the previous char was a variable

-or-

if the current char is x,

then if the next char is ^ … this is how I checked for squares before… What is the difference?

I do not think there is a difference sufficient to determine one method superior to the other, so we will follow with checking ^

so that the current check for squares can remain in place, and the code is less altered that way… we will see if it bites me in the ass.

So,

if the previous char was a variable

then this is the term we are looking for. exponent = degree

if the character two steps back (behind the variable) is a digit,

wait this number was already parsed in X! let's not do it again. Therefore, I was wrong! (figured it out before it bit me though)

So, scratch that.

if the current character is x

if the next char is ^

the number in the string = coefficient

add this to an array to keep track of all of them.

the number after the ^ (two steps ahead of the current char) = the degree

At this point, we have the root, coefficients, and degrees of the equation.

Relevant fundamental rules of algebra:

for every polynomial of degree n, there are exactly n roots.

Abel Ruffini Theorem: there is no general formula for finding roots of polynomials of degree 5+ using only arithmetic.

…so, no equations above 5 degrees. okay.

A 4-degree polynomial in standard form has at most five terms.

So, assign variables for five terms, set equal to null.

if term1 == null

if term2 == null

if term3 == null?

No, use the exponent! x = term1 x^2 = term2 and so on

then with all the terms, determine the degree

if term 4 != null, degree =4

else if term 3!= null, degree =3

and so on.

if the degree is <= 2, use existing solve x
else if the degree is 3 or 4, use rational roots
else if the degree is higher than for reject for Abel Ruffini theorem.
I guess reject for invalid input, I guess?

for 3 and 4 degrees, generate rational root candidates from term variables,
evaluate each with rpn eval
   for each root found,
      record (x +/- root)
      perform division to reduce polynomial
   if quadratic remains
      use quad
   if no rational roots found
      error prime polynomial
   display factored form

So, all I need to do is update the variable place and add a function to manage the terms. rpn and quad logic remain untouched.

let’s try it!